

KEWAUNEE, WI, USA

WMO: 997741

Lat: 44.4650N

Lon: 87.4960W

Elev: 178

StdP: 99.20

Time Zone: -6.00 (NAC)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.9	-18.3	N/A	N/A	N/A	N/A	N/A	N/A	13.2	-1.9	11.9	-2.6	4.8	290	0.599

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.0	28.0	N/A	25.7	N/A	23.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	4.3	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.2	9.5	8.5	-23.7	31.3	3.5	2.2	-26.2	32.9	-28.3	34.2	-30.3	35.4	-32.8	37.0		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.6	-7.2	-6.5	-0.9	4.9	10.3	15.1	19.0	19.0	15.1	9.4	2.6	-3.2
	DBStd	10.15	6.44	5.88	5.39	3.47	3.44	3.43	3.25	2.59	3.47	3.65	4.63	5.44
	HDD10.0	2218	534	461	337	156	38	2	0	0	3	57	222	408
	HDD18.3	4416	792	694	595	402	249	108	31	22	108	278	471	667
	CDD10.0	961	0	0	1	4	48	156	279	279	154	37	1	0
	CDD18.3	116	0	0	0	0	0	12	52	43	9	0	0	0
	CDH23.3	492	0	0	0	1	9	75	252	122	31	2	0	0
	CDH26.7	113	0	0	0	0	2	13	71	21	7	0	0	0
Wind	WSAvg	4.0	4.6	4.6	4.4	4.7	4.2	3.3	3.1	3.1	3.5	4.1	4.4	4.4
Precipitation	PrecAvg	781	39	28	46	71	86	99	91	80	76	67	54	44
	PrecMax	939	79	54	100	125	188	218	172	148	139	148	132	123
	PrecMin	654	12	7	8	23	40	34	31	23	28	15	9	10
	PrecStd	69	20	12	26	28	34	49	40	28	27	35	29	25
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.2	6.7	13.6	18.2	25.1	29.0	31.8	29.1	27.1	21.1	13.8	8.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	4.7	3.7	10.0	13.8	20.4	25.9	28.7	26.5	23.4	17.2	11.0	6.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	3.2	2.5	7.9	10.9	17.7	23.0	26.3	24.4	21.6	15.7	9.7	5.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	1.9	1.3	5.8	9.1	15.1	20.3	23.9	23.2	20.3	14.7	8.5	3.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	7.0	7.3	7.1	6.1	7.0	6.7	7.4	7.0	7.3	6.3	5.7	6.0
		MCDBR	7.7	6.6	8.7	9.4	10.9	11.4	10.7	8.7	9.1	8.0	6.4	6.1
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.276	0.302	0.330	0.377	0.402	0.413	0.404	0.400	0.378	0.337	0.307	0.276	
	taud	2.434	2.345	2.387	2.304	2.289	2.301	2.336	2.343	2.395	2.500	2.514	2.488	
	Ebn at Noon	855	886	907	885	869	858	861	852	844	840	813	823	
	Edh at Noon	71	94	104	123	129	128	122	117	103	81	67	62	
All-Sky Solar Radiation	RadAvg	1.47	2.46	3.61	4.47	5.42	5.91	6.05	5.17	4.07	2.55	1.58	1.14	
	RadStd	0.18	0.25	0.32	0.43	0.44	0.43	0.25	0.34	0.32	0.25	0.15	0.12	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page